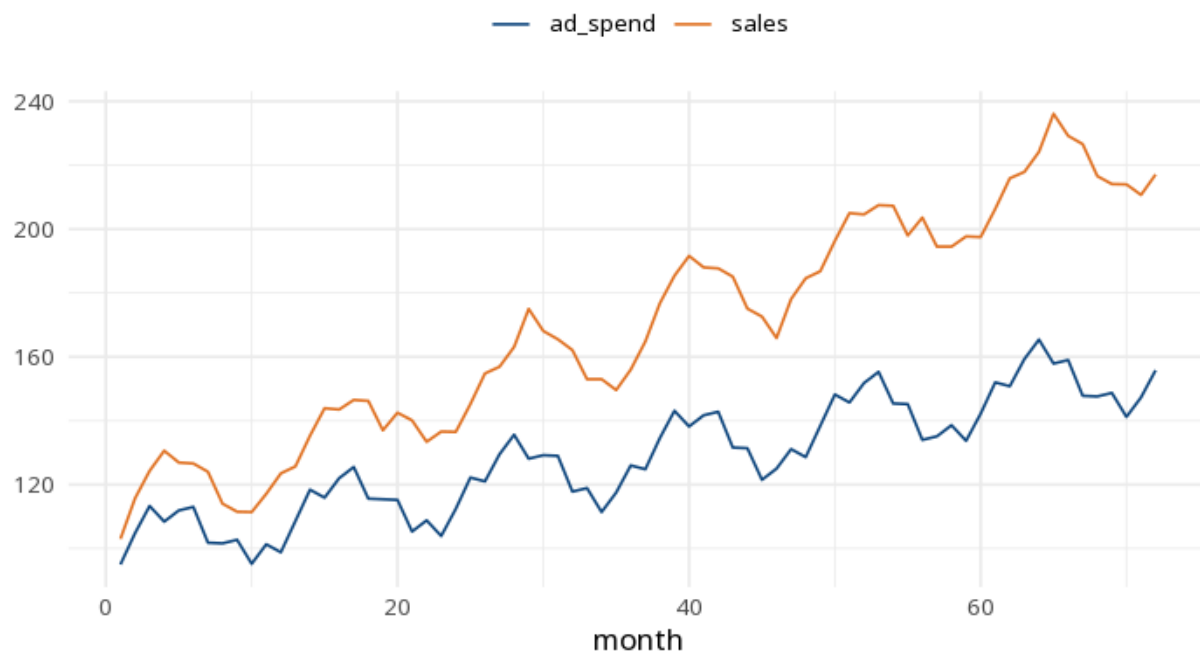


1 Granger causality

Direction	F	df	p
X→Y	47.89	4, 63	<.001
Y→X	30.62	4, 63	<.001

Lag	AIC	BIC	HQ
1	6.78	6.98	6.86
2	5.80	6.12	5.93
3	5.36	5.81	5.54
4	4.96	5.55	5.19



A significant $X \rightarrow Y$ p means past values of X help predict Y beyond Y's own past — this is predictive precedence, not proof that X causes Y. Check both directions. The result depends entirely on the lag order (this card uses a max lag of 4): choose it on theory or by an information criterion — `vars::VARselect` reports the AIC- and BIC-minimising lag — and report it, since different lags can flip the conclusion. Make both series stationary first. Method: `R lmtest::grangertest` (Granger, 1969). Your significance threshold (α) is 0.05.

Granger test $X \rightarrow Y$: $F(4,63)=47.89$, $p < .001$; $Y \rightarrow X$: $F(4,63)=30.62$, $p < .001$ (lag=4).

Statistical basis: Granger causality F-test (Granger, C. W. J. (1969). "Investigating causal relations by econometric models and cross-spectral methods."

Econometrica, 37(3), 424-438.); Implementation (`lmtest::grangertest`) (Zeileis A, Hothorn T (2002). "Diagnostic Checking in Regression Relationships." R News, 2(3), 7-10.).